LaTeX Table Practice Pack

10 Graded Exercises Modelled on the Lab-Exam Papers

Reproduce Part I from scratch — check Part II afterwards

July 21, 2026

Rules of engagement. Print Part I only. Start from a blank file with your own preamble. Time each attempt. A table "passes" only if it *compiles* and matches the target structurally — merged cells in the right places, rules in the right places, maths rendered as maths.

Contents

PART I — Target Tables

Reproduce each of the following

Exercise 1 — Warm-up

Difficulty: $\star$ **Target time:** 3 minutes
Tests: column spec, borders, inline maths in cells

Table 1: Sorting-algorithm complexity.

Algorithm	**Best**	**Average**	**Worst**
Bubble sort	$\mathcal{O}(n)$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$
Merge sort	$\mathcal{O}(n \log n)$	$\mathcal{O}(n \log n)$	$\mathcal{O}(n \log n)$
Quick sort	$\mathcal{O}(n \log n)$	$\mathcal{O}(n \log n)$	$\mathcal{O}(n^2)$
Binary search	$\mathcal{O}(1)$	$\mathcal{O}(\log n)$	$\mathcal{O}(\log n)$

Exercise 2 — Two-tier header

Difficulty: $\star\star$ **Target time:** 5 minutes
Tests: `multicolumn`, `multirow`, `cline` — the header pattern from the Laplace paper

Table 2: Numerical integration methods.

Method	**Error**		**Order**
	Local	**Global**	
Trapezoidal	$\mathcal{O}(h^3)$	$\mathcal{O}(h^2)$	2
Simpson	$\mathcal{O}(h^5)$	$\mathcal{O}(h^4)$	4
Euler	$\mathcal{O}(h^2)$	$\mathcal{O}(h)$	1

Exercise 3 — Multirow families with stacked fractions

Difficulty: $\star\star\star$ **Target time:** 8 minutes
Tests: `multirow` spanning 3 rows, `dfrac` in cells, `arraystretch`, partial `cline`

Table 3: Standard derivatives and integrals by family.

Family	**Operation**		**Domain**
	Function	**Derivative**	
Power	x^n	nx^{n-1}	$n \in \mathbb{R}$
	$\frac{1}{x}$	$-\frac{1}{x^2}$	$x \neq 0$
	$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	$x > 0$
Trigonometric	$\sin x$	$\cos x$	$x \in \mathbb{R}$
	$\tan x$	$\sec^2 x$	$x \neq \frac{\pi}{2}$
Exponential	e^{ax}	ae^{ax}	$a \in \mathbb{R}$
	$\ln x$	$\frac{1}{x}$	$x > 0$
Chain rule	$f(g(x))$	$f'(g(x))\, g'(x)$	differentiable

Exercise 4 — Coloured table with scientific notation

Difficulty: ⋆⋆⋆ **Target time:** 8 minutes
Tests: `rowcolor`, `cellcolor`, fraction inside a header cell, $\times 10^{-n}$ notation

Table 4: Semiconductor material parameters.

Material	**Electrical**		**Derived**	
	E_g (eV)	μ_n	$\sigma = \dfrac{1}{\rho}$	$\beta = \sqrt{\mu_n}$
Silicon	1.12	1.35×10^3	4.35×10^{-4}	36.7
Germanium	0.67	3.90×10^3	2.17×10^{-2}	62.4
GaAs	1.42	8.50×10^3	1.00×10^{-8}	92.2
InP	1.34	5.40×10^3	1.25×10^{-7}	73.5
		Mean gap	1.14 eV	

Exercise 5 — Spanning in both directions

Difficulty: ⋆⋆⋆ **Target time:** 7 minutes
Tests: `multirow` and `multicolumn` interacting, merged footer row

Table 5: Course assessment breakdown.

Component	**Item**	**Weight**	**Marks**
Continuous	Quizzes	10%	10
	Assignments	15%	15
	Lab work	15%	15
Examination	Midterm	25%	25
	Final	35%	35
Total		**100%**	**100**

Exercise 6 — Professional rules with wrapped cell text

Difficulty: ⋆⋆ **Target time:** 6 minutes
Tests: `booktabs`, `makecell` line breaks inside a cell, no vertical rules

Table 6: Comparison of transform methods.

Transform	**Kernel function**	**Typical application**
Laplace	e^{-st}	transient circuits, control systems
Fourier	$e^{-j\omega t}$	spectra, signal filtering
Z	z^{-n}	discrete systems, digital filters
All three convert differential relations into algebraic ones.		

Exercise 7 — Fixed total width with a wrapping column

Difficulty: ⋆⋆ **Target time:** 5 minutes
Tests: `tabularx`, the `X` column, maths inside a paragraph column

Table 7: Boundary conditions and their meaning.

Name	**Form**	**Physical interpretation**
Dirichlet	$u = g$ on $\partial\Omega$	The value of the solution is prescribed on the boundary, for example a surface held at a fixed temperature.
Neumann	$\dfrac{\partial u}{\partial n} = g$	The flux across the boundary is prescribed; a zero value models a perfectly insulated wall.
Robin	$\alpha u + \beta \dfrac{\partial u}{\partial n} = g$	A weighted combination of the two above, used for convective heat exchange with a surrounding medium.

Exercise 8 — Heavy maths inside cells

Difficulty: ⋆⋆⋆⋆ **Target time:** 10 minutes
Tests: matrices, `cases`, integrals and limits nested inside table cells

Table 8: Objects that must survive inside a cell.

Object	**Expression**	**Note**
Matrix	$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$	2×2 real
Determinant	$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$	identity
Piecewise	$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$	continuous at 0
Integral	$\displaystyle\int_0^\infty e^{-st}\,dt = \frac{1}{s}$	$s > 0$
Limit	$\displaystyle\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$	classical
Nested index	$\displaystyle\sum_{\substack{i=1 \\ i\neq j}}^{n} a_{i_j}$	double subscript

Exercise 9 — Three-tier header with a diagonal cell

Difficulty: ⋆⋆⋆⋆⋆ **Target time:** 12 minutes
Tests: `diagbox`, a `multirow` spanning three header rows, repeated `multicolumn` tiers

Table 9: Network layer performance matrix.

Metric / Layer	**Training**		**Inference**	
	(per epoch)		(per sample)	
	Time	**Memory**	**Time**	**Memory**
Convolution	12.4 s	1.8 GB	3.1 ms	0.4 GB
Pooling	0.8 s	0.2 GB	0.2 ms	0.1 GB
Dense	4.6 s	3.4 GB	1.0 ms	0.9 GB
Total	17.8 s	5.4 GB	4.3 ms	1.4 GB

Exercise 10 — Full exam simulation

Difficulty: ⋆⋆⋆⋆⋆ **Target time:** 15 minutes
Tests: everything at once — colour, both spans, nested maths, footnote marker, caption, label and a cross-reference from the body text

Table **??** summarises the four candidate designs.

Table 10: Design trade-off study for the RC filter stage.

Design	**Components**		**Response**	
	R (kΩ)	C (μF)	$\tau = RC$	$f_c = \dfrac{1}{2\pi RC}$
Fast	1.0	0.10	0.10 ms	1.59 kHz
	2.2	0.10	0.22 ms	723 Hz
Slow	10.0	1.00	10.0 ms	15.9 Hz
	22.0	4.70	103 ms	1.54 Hz[1]
Selected design			Fast, $R = 1.0$ kΩ	

The selected design in Table **??** gives the widest bandwidth.

[1] Measured with a 5 V step input at 25°C.

PART II — Solutions and Marking Notes

Do not read until you have attempted Part I

Solution 1

```
\begin{table}[H]
\centering
\caption{Sorting-algorithm complexity.}
\begin{tabular}{|l|c|c|c|}
\hline
\textbf{Algorithm} & \textbf{Best} & \textbf{Average} & \textbf{Worst} \\
\hline
Bubble sort   & $\mathcal{O}(n)$       & $\mathcal{O}(n^2)$     & $\mathcal{O}(n^2)$ \\
\hline
Merge sort    & $\mathcal{O}(n\log n)$ & $\mathcal{O}(n\log n)$ & $\mathcal{O}(n\log n)$ \\
\hline
Quick sort    & $\mathcal{O}(n\log n)$ & $\mathcal{O}(n\log n)$ & $\mathcal{O}(n^2)$ \\
\hline
Binary search & $\mathcal{O}(1)$       & $\mathcal{O}(\log n)$  & $\mathcal{O}(\log n)$ \\
\hline
\end{tabular}
\end{table}
```

Marking notes. The column spec `{|l|c|c|c|}` has four entries and four `|` fences — count them before typing the body. `\log` must be a command, not the letters `log`, or it prints in italic. `\mathcal{O}` needs `amssymb`/`amsfonts`.

Solution 2

```
\begin{table}[H]
\centering
\caption{Numerical integration methods.}
\renewcommand{\arraystretch}{1.3}
\begin{tabular}{|l|c|c|c|}
\hline
\multirow{2}{*}{\textbf{Method}} & \multicolumn{2}{c|}{\textbf{Error}} &
  \multirow{2}{*}{\textbf{Order}} \\
\cline{2-3}
 & \textbf{Local} & \textbf{Global} & \\
\hline
Trapezoidal & $\mathcal{O}(h^3)$ & $\mathcal{O}(h^2)$ & 2 \\
\hline
Simpson     & $\mathcal{O}(h^5)$ & $\mathcal{O}(h^4)$ & 4 \\
\hline
Euler       & $\mathcal{O}(h^2)$ & $\mathcal{O}(h)$   & 1 \\
\hline
\end{tabular}
\end{table}
```

Marking notes. The second header row starts with a bare `&` and ends with `& \\` — those two cells are *already occupied* by the `\multirow` above. Forgetting the empty cells is the single most common failure. `\cline{2-3}` rules only the two Error columns; a full `\hline` there would cut through “Method” and “Order”.

Solution 3

```latex
\begin{table}[H]
\centering
\caption{Standard derivatives and integrals by family.}
\renewcommand{\arraystretch}{1.7}
\begin{tabular}{|c|c|c|c|}
\hline
\multirow{2}{*}{\textbf{Family}} &
  \multicolumn{2}{c|}{\textbf{Operation}} &
  \multirow{2}{*}{\textbf{Domain}} \\
\cline{2-3}
 & \textbf{Function} & \textbf{Derivative} & \\
\hline
\multirow{3}{*}{Power}
  & $x^n$          & $n x^{n-1}$            & $n \in \mathbb{R}$ \\ \cline{2-4}
  & $\dfrac{1}{x}$ & $-\dfrac{1}{x^2}$      & $x \neq 0$ \\ \cline{2-4}
  & $\sqrt{x}$     & $\dfrac{1}{2\sqrt{x}}$ & $x > 0$ \\
\hline
\multirow{2}{*}{Trigonometric}
  & $\sin x$ & $\cos x$    & $x \in \mathbb{R}$ \\ \cline{2-4}
  & $\tan x$ & $\sec^2 x$  & $x \neq \dfrac{\pi}{2}$ \\
\hline
\multirow{2}{*}{Exponential}
  & $e^{ax}$ & $a e^{ax}$     & $a \in \mathbb{R}$ \\ \cline{2-4}
  & $\ln x$  & $\dfrac{1}{x}$ & $x > 0$ \\
\hline
\multicolumn{2}{|c|}{\textbf{Chain rule } $f(g(x))$}
  & $f'(g(x))\,g'(x)$ & differentiable \\
\hline
\end{tabular}
\end{table}
```

Marking notes. This is the Laplace-table pattern with different content. Three rules to internalise: (i) after a `\multirow{3}`, the next *two* rows must begin with a bare `&`; (ii) use `\cline{2-4}` between sub-rows so the family label stays unbroken, and a full `\hline` only when the family changes; (iii) without `\renewcommand{\arraystretch}{1.7}` the stacked `\dfracs` collide. The last row flips orientation — `\multicolumn{2}{c|...|` merges *across* instead of down.

Solution 4

```latex
\begin{table}[H]
\centering
\caption{Semiconductor material parameters.}
\renewcommand{\arraystretch}{1.5}
\begin{tabular}{|l|cc|cc|}
\hline
\rowcolor{Gray!35}
\multirow{2}{*}{\textbf{Material}} &
  \multicolumn{2}{c|}{\textbf{Electrical}} &
  \multicolumn{2}{c|}{\textbf{Derived}} \\
\cline{2-5}
\rowcolor{Gray!20}
 & $E_g$ (eV) & $\mu_n$ & $\sigma = \dfrac{1}{\rho}$ & $\beta = \sqrt{\mu_n}$ \\
\hline
\rowcolor{Yellow!15} Silicon   & 1.12 & $1.35\times10^{3}$ & $4.35\times10^{-4}$ & $36.7$ \\
\rowcolor{Yellow!25} Germanium & 0.67 & $3.90\times10^{3}$ & $2.17\times10^{-2}$ & $62.4$ \\
\rowcolor{Cyan!15}   GaAs      & 1.42 & $8.50\times10^{3}$ & $1.00\times10^{-8}$ & $92.2$ \\
\rowcolor{Cyan!25}   InP       & 1.34 & $5.40\times10^{3}$ & $1.25\times10^{-7}$ & $73.5$ \\
\hline
\multicolumn{3}{|r|}{\cellcolor{Green!15}\textbf{Mean gap}} &
  \multicolumn{2}{c|}{\cellcolor{Green!15}$1.14$ eV} \\
```

```
\hline
\end{tabular}
\end{table}
```

Marking notes. `\rowcolor` must be the **first token of the row**, before `\multirow` — anywhere else and it errors. The colour package must be loaded as `\usepackage[table]{xcolor}`; without the `[table]` option `\rowcolor` is undefined. `Gray!35` means 35 % Gray on white. Note two `\multicolumn`s coexist in the final row, each carrying its own `\cellcolor`.

Solution 5

```
\begin{table}[H]
\centering
\caption{Course assessment breakdown.}
\renewcommand{\arraystretch}{1.4}
\begin{tabular}{|c|l|c|c|}
\hline
\textbf{Component} & \textbf{Item} & \textbf{Weight} & \textbf{Marks} \\
\hline
\multirow{3}{*}{Continuous}
  & Quizzes     & 10\% & 10 \\ \cline{2-4}
  & Assignments & 15\% & 15 \\ \cline{2-4}
  & Lab work    & 15\% & 15 \\
\hline
\multirow{2}{*}{Examination}
  & Midterm & 25\% & 25 \\ \cline{2-4}
  & Final   & 35\% & 35 \\
\hline
\multicolumn{2}{|c|}{\textbf{Total}} & \textbf{100\%} & \textbf{100} \\
\hline
\end{tabular}
\end{table}
```

Marking notes. The percent sign must be escaped as `\%` — a bare `%` comments out the rest of the line and silently eats your row. This is a favourite way to lose marks. The footer merges the first two columns while the last two stay separate.

Solution 6

```
\begin{table}[H]
\centering
\caption{Comparison of transform methods.}
\renewcommand{\arraystretch}{1.2}
\begin{tabular}{lcc}
\toprule
\textbf{Transform} & \makecell{\textbf{Kernel}\\\textbf{function}} &
  \makecell{\textbf{Typical}\\\textbf{application}} \\
\midrule
Laplace  & $e^{-st}$        & \makecell{transient circuits,\\control systems} \\
Fourier  & $e^{-j\omega t}$ & \makecell{spectra,\\signal filtering} \\
Z        & $z^{-n}$         & \makecell{discrete systems,\\digital filters} \\
\midrule
\multicolumn{3}{c}{\textit{All three convert differential relations into algebraic ones.}} \\
\bottomrule
\end{tabular}
\end{table}
```

Marking notes. `booktabs` style deliberately has *no* vertical rules, so the column spec is plain `{lcc}`. Never mix `\hline` with `\toprule`/`\midrule`. `\makecell{a\\b}` is how you break a line inside a cell — a bare `\\` there would end the whole row instead.

Solution 7

```
\begin{table}[H]
\centering
\caption{Boundary conditions and their meaning.}
\renewcommand{\arraystretch}{1.3}
\begin{tabularx}{0.92\textwidth}{|l|c|X|}
\hline
\textbf{Name} & \textbf{Form} & \textbf{Physical interpretation} \\
\hline
Dirichlet & $u = g$ on $\partial\Omega$ &
  The value of the solution is prescribed on the boundary, for example a
  surface held at a fixed temperature. \\
\hline
Neumann & $\dfrac{\partial u}{\partial n} = g$ &
  The flux across the boundary is prescribed; a zero value models a
  perfectly insulated wall. \\
\hline
Robin & $\alpha u + \beta \dfrac{\partial u}{\partial n} = g$ &
  A weighted combination of the two above, used for convective heat
  exchange with a surrounding medium. \\
\hline
\end{tabularx}
\end{table}
```

Marking notes. `tabularx` takes an *extra* first argument: the total width. The `X` column absorbs the leftover space and wraps text automatically — this is the fix whenever a table overflows the margin. You may use several `X` columns; they share the space equally.

Solution 8

```
\begin{table}[H]
\centering
\caption{Objects that must survive inside a cell.}
\renewcommand{\arraystretch}{2.0}
\begin{tabular}{|c|c|c|}
\hline
\textbf{Object} & \textbf{Expression} & \textbf{Note} \\
\hline
Matrix & $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$
       & $2 \times 2$ real \\
\hline
Determinant & $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$ & identity \\
\hline
Piecewise & $f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$
          & continuous at $0$ \\
\hline
Integral & $\displaystyle\int_{0}^{\infty} e^{-st} \, dt = \dfrac{1}{s}$
         & $s > 0$ \\
\hline
Limit & $\displaystyle\lim_{n \to \infty} \left(1 + \dfrac{1}{n}\right)^{n} = e$
      & classical \\
\hline
Nested index & $\displaystyle\sum_{\substack{i=1 \\ i \neq j}}^{n} a_{i_{j}}$
             & double subscript \\
```

```
\hline
\end{tabular}
\end{table}
```

Marking notes. The surprise here is that `&` and `\\` inside `bmatrix`/`cases` do *not* break the outer table — the inner environment claims them. That is why the matrix row works despite containing three `&` and one `\\`. Add `\displaystyle` to force full-size $\sum$, $\int$ and lim with limits above and below; without it they shrink to inline size. Push `\arraystretch` up to 2.0 to stop tall cells colliding.

Solution 9

```
\begin{table}[H]
\centering
\caption{Network layer performance matrix.}
\renewcommand{\arraystretch}{1.4}
\begin{tabular}{|c|c|c|c|c|}
\hline
\multirow{3}{*}{\diagbox[width=6em]{\textbf{Layer}}{\textbf{Metric}}} &
  \multicolumn{2}{c|}{\textbf{Training}} &
  \multicolumn{2}{c|}{\textbf{Inference}} \\
\cline{2-5}
 & \multicolumn{2}{c|}{(per epoch)} & \multicolumn{2}{c|}{(per sample)} \\
\cline{2-5}
 & \textbf{Time} & \textbf{Memory} & \textbf{Time} & \textbf{Memory} \\
\hline
Convolution & $12.4$ s & $1.8$ GB & $3.1$ ms & $0.4$ GB \\
\hline
Pooling     & $0.8$ s  & $0.2$ GB & $0.2$ ms & $0.1$ GB \\
\hline
Dense       & $4.6$ s  & $3.4$ GB & $1.0$ ms & $0.9$ GB \\
\hline
\multicolumn{1}{|r|}{\textbf{Total}} & $17.8$ s & $5.4$ GB & $4.3$ ms & $1.4$ GB \\
\hline
\end{tabular}
\end{table}
```

Marking notes. A `\multirow{3}` means the next *two* rows each open with a bare `&`. `\diagbox{A}{B}` (package `diagbox`) splits one cell with a diagonal; fix its width with the optional `[width=6em]` or the column will bulge. If `diagbox` is unavailable on the exam machine, substitute `\multirow{3}{*}{\textbf{L` and move on — structure earns more marks than the diagonal. The last row uses `\multicolumn{1}{r|...|` purely to *re-align* a single cell to the right, which is a legitimate and handy trick.

Solution 10

```
Table~\ref{tab:sim} summarises the four candidate designs.

\begin{table}[H]
\centering
\caption{Design trade-off study for the RC filter stage.}
\label{tab:sim}
\renewcommand{\arraystretch}{1.6}
\begin{tabular}{|l|c|c|c|c|}
\hline
\rowcolor{RoyalBlue!25}
\multirow{2}{*}{\textbf{Design}} &
  \multicolumn{2}{c|}{\textbf{Components}} &
  \multicolumn{2}{c|}{\textbf{Response}} \\
```

```
\cline{2-5}
\rowcolor{RoyalBlue!12}
 & $R$ (k$\Omega$) & $C$ ($\mu$F) &
   $\tau = RC$ & $f_c = \dfrac{1}{2\pi RC}$ \\
\hline
\multirow{2}{*}{Fast}
  & 1.0 & 0.10 & $0.10$ ms & $1.59$ kHz \\ \cline{2-5}
  & 2.2 & 0.10 & $0.22$ ms & $723$ Hz \\
\hline
\multirow{2}{*}{Slow}
  & 10.0 & 1.00 & $10.0$ ms & $15.9$ Hz \\ \cline{2-5}
  & 22.0 & 4.70 & $103$ ms  & $1.54$ Hz\footnotemark \\
\hline
\rowcolor{Green!12}
\multicolumn{3}{|r|}{\textbf{Selected design}} &
  \multicolumn{2}{c|}{Fast, $R = 1.0$ k$\Omega$} \\
\hline
\end{tabular}
\end{table}
\footnotetext{Measured with a $5$\,V step input at $25^{\circ}$C.}

The selected design in Table~\ref{tab:sim} gives the widest bandwidth.
```

Marking notes. Five things are graded at once here. `\caption` goes *before* `\label`, and `\label` must follow `\caption` or `\ref` prints the wrong number. `\ref{tab:sim}` resolves only on the **second** compile — a `??` means run again, not that your code is wrong. A normal `\footnote` is swallowed inside a float, so use `\footnotemark` in the cell and `\footnotetext` just after the table. Greek and units inside cells are ordinary maths: `k$\Omega$`, `$\mu$F`, `$25^{\circ}$C`. And `\rowcolor` still has to lead its row even when a `\multicolumn` follows.

Drill Routine

1. **Round 1 — untimed.** Work through 1–5 with this file's Part II open. The aim is recognition, not speed.
2. **Round 2 — timed, easy half.** Close Part II. Reproduce 1, 2, 5, 6 in under 20 minutes total. If you exceed it, the bottleneck is almost always counting columns — write the column spec first and count the `|`.
3. **Round 3 — timed, hard half.** Reproduce 3, 4, 8 in under 25 minutes. These three contain every pattern the real papers used.
4. **Round 4 — exam pace.** Do Exercise 10 cold in 15 minutes, including the surrounding sentence and the cross-reference.
5. **Round 5 — from memory.** Write Exercise 3 with no reference open at all. When you can do that, tables will not cost you marks.

The Five Errors That Cost the Most Marks

#	Mistake	Symptom and fix
1	Missing `&` after a `\multirow`	`Extra alignment tab` or shifted columns. Every row covered by a `\multirow {N}` needs a placeholder `&` in that column.
2	`\rowcolor` not first	Error or uncoloured row. It must precede `\multirow` and `\multicolumn`.
3	Bare `A row silently vanishes. Escape them: \verb`%| and `\&`.	
4	Column count mismatch	`Extra alignment tab has been changed to \cr`. Count the `&` in the row: it must be one fewer than the number of columns, after accounting for spans.
5	Compiling once	`\ref` shows `??` and the TOC is stale. Always run `pdflatex` twice.

Reproduce, compile, compare, repeat.